

Artificial Intelligence Clones

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Sloan-Nomis

introduction

LLMs, fine-tuned on our personal data, may soon be able to mimic individual personalities (“AI clones,” “digital twins,” “AI agents”)

**AI Agents Simulate 1,052 Individuals’
Personalities with Impressive
Accuracy**

Stanford HCAI, 2025

It’s no longer science fiction: Your personality – your beliefs, quirks, and decision-making patterns – can be captured and brought to life inside an artificial mind.

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Create Your Digital Self

Your Mind. Your Model. Your Rules.
Answer Questions & Connect 24/7 In Your Voice

Create My Delphi

Explore Experts

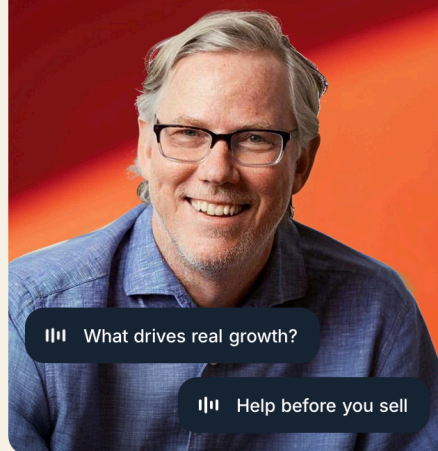
Lenny Rachitsky
Product Expert



Sam Feldt
Musician



Brian Halligan
Executive



Vanessa Van Edwards
Behavioral Scientist



Jay Shetty
Author & Podcast Host



introduction

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this technology has the potential to transform how we search over human candidates, in particular in settings where direct evaluation is costly

labor and dating markets are already moving in this direction

REPLICANT

AI-powered candidate replication platform helping job seekers create intelligent digital representatives for recruitment conversations.



AI Training

Upload your resume and provide context about your experience, goals, and preferences



Clone Creation

Our AI creates an intelligent representation that understands your professional profile



Smart Matching

Your AI clone engages with recruiters, answering questions and showcasing your skills

Ready to Get Started?

Join the future of recruitment and let your AI clone work for you.

Create Your AI Clone

Ribbon raised \$8.2M to fix hiring globally >

Meet your AI-powered recruiter

Ribbon talks with every candidate, scores their answers, and shares insights that help you hire faster and smarter.

Book a demo

Start for free



Daphne D.
Operations Manager

This was my first time doing an interview like this and it was so smooth!



Hassan J.
DevOps Engineer

I like the new AI approach, and I thought the agent was very interactive and smart and responded well to changing dynamics of the conversation!



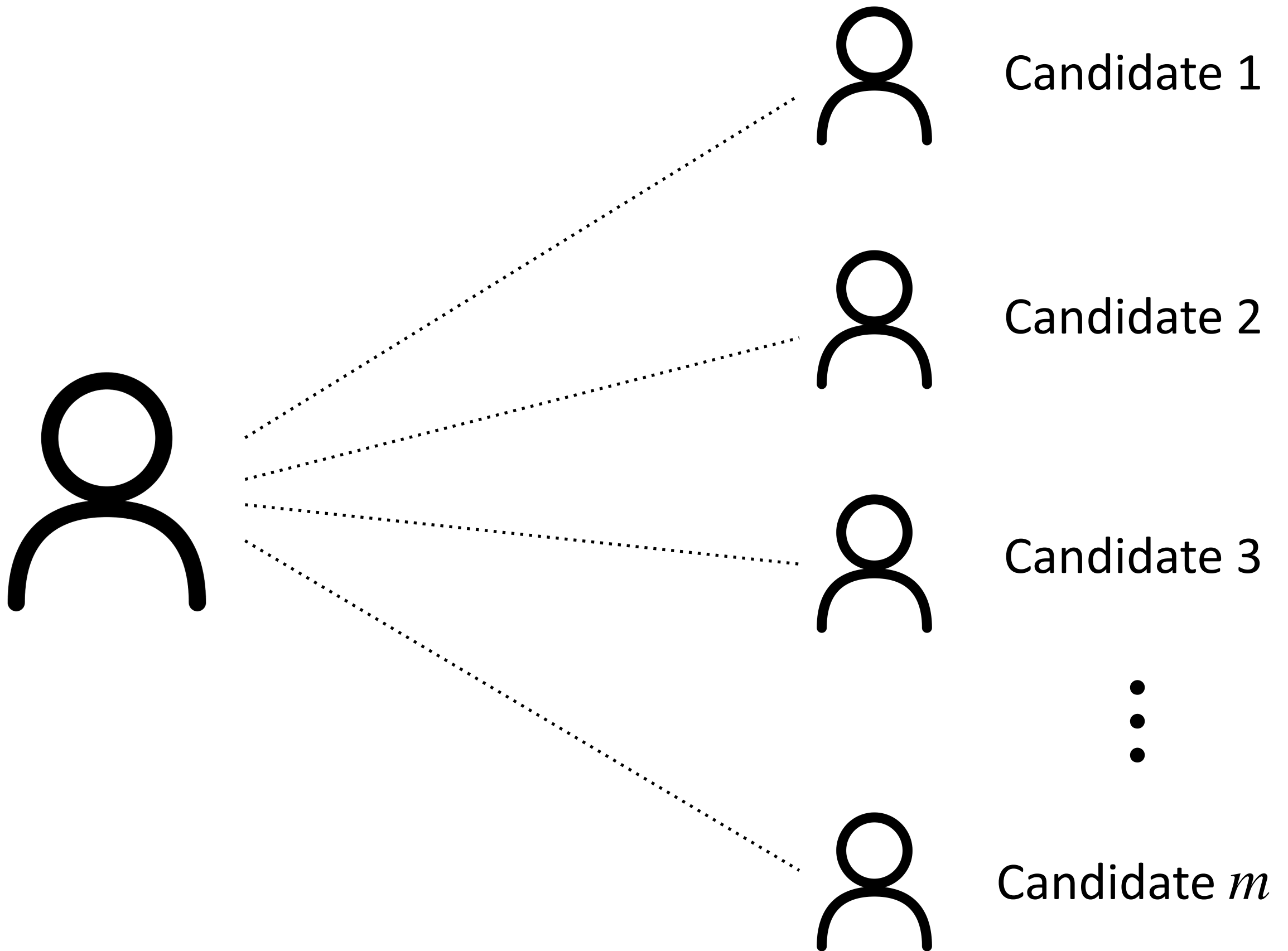
Jamie K.

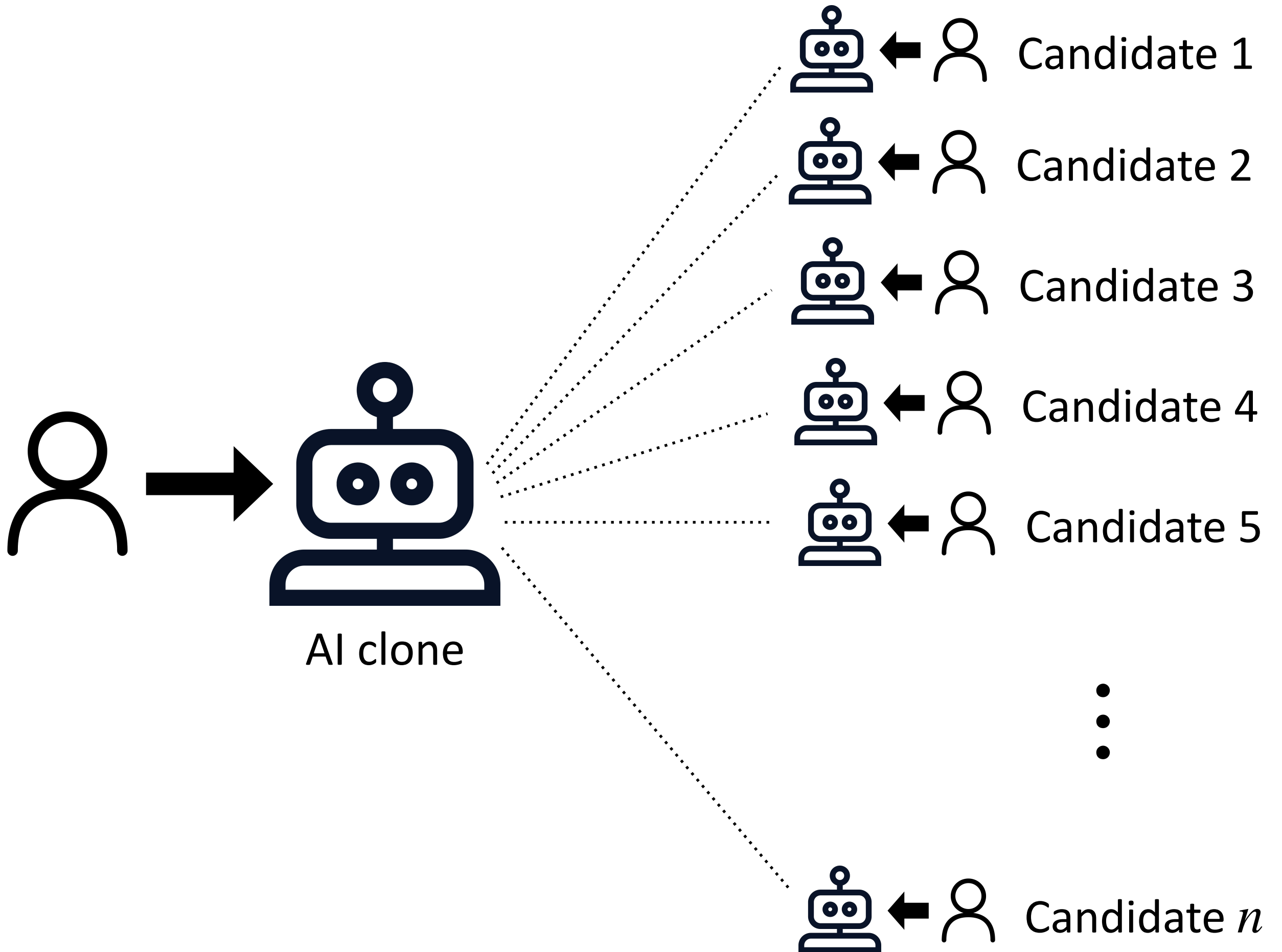
Are A.I. Clones the Future of Dating? I Tried Them for Myself.

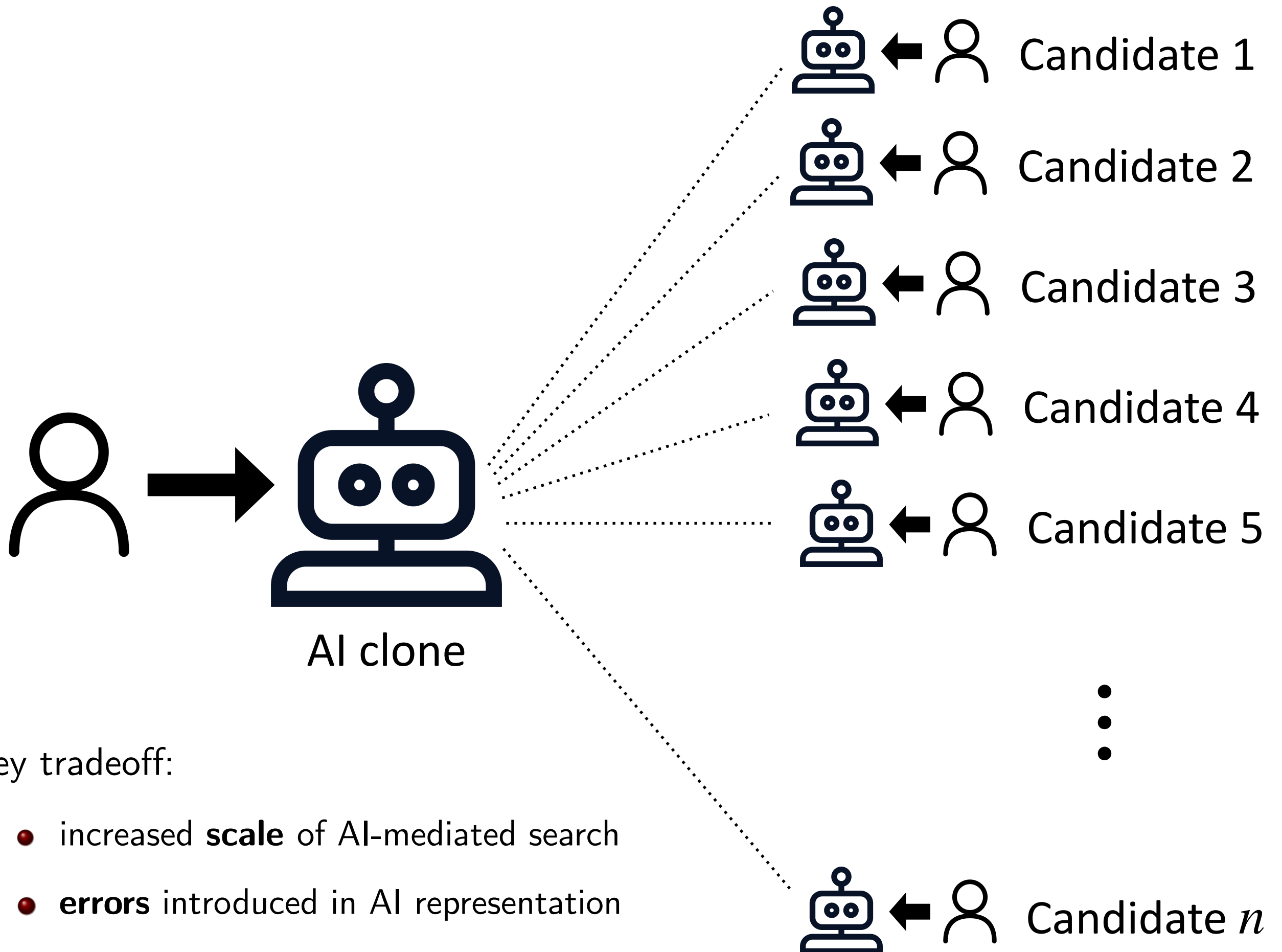
executives in the dating app industry have begun pitching a future in which people can create A.I. clones of themselves that date other clones and relay the results back to their human counterparts.

Dating Apps Suck. A.I. Clones Are Making Them Even Weirder.

‘My clone is on its way to date all of San Francisco in one month.’







Model

Setup

Individuals are represented by vectors in $\mathbb{R}^k$

- each entry is an attribute of the person

possible attributes:

extraversion

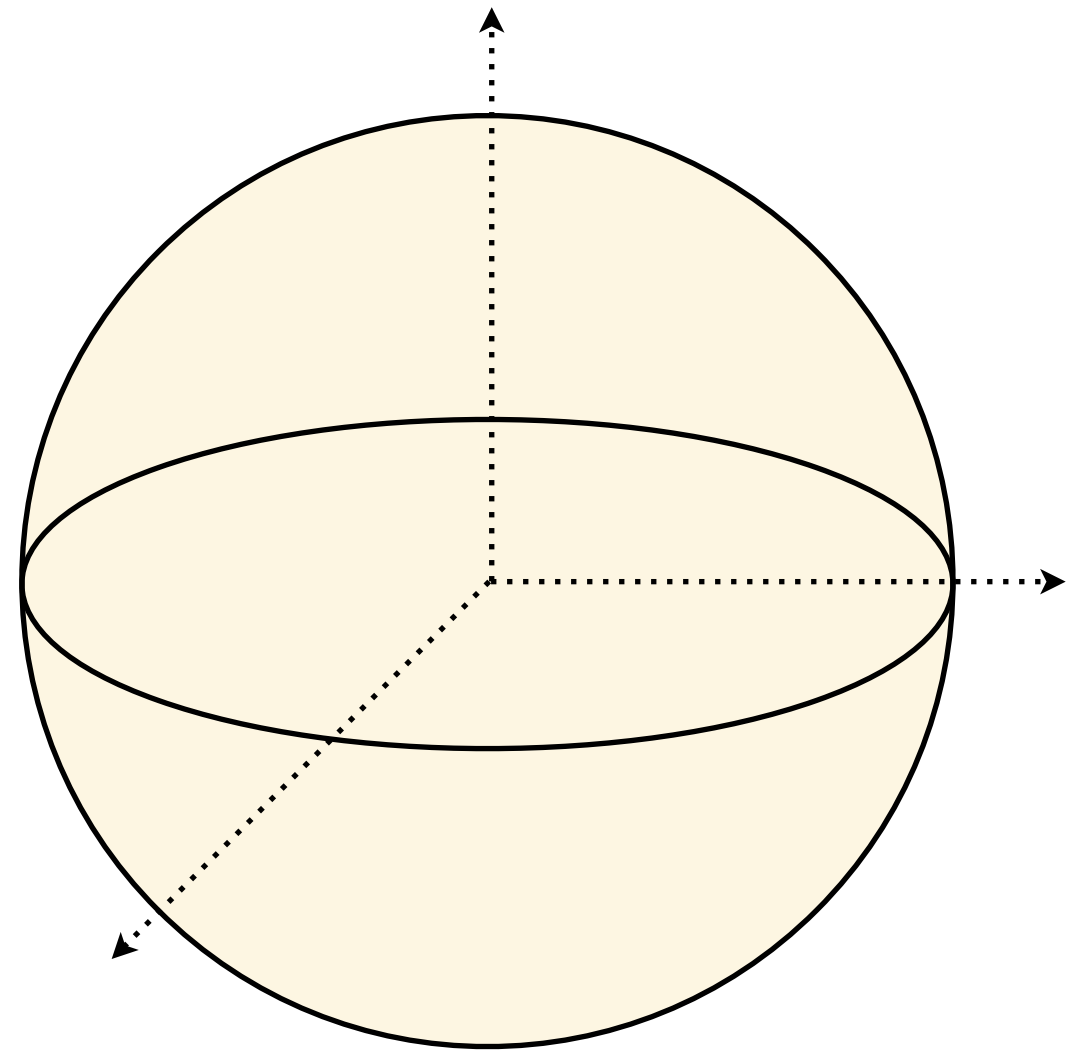
intellectual curiosity

relationship to time

style of humor

relationship to authority

speed of decision-making

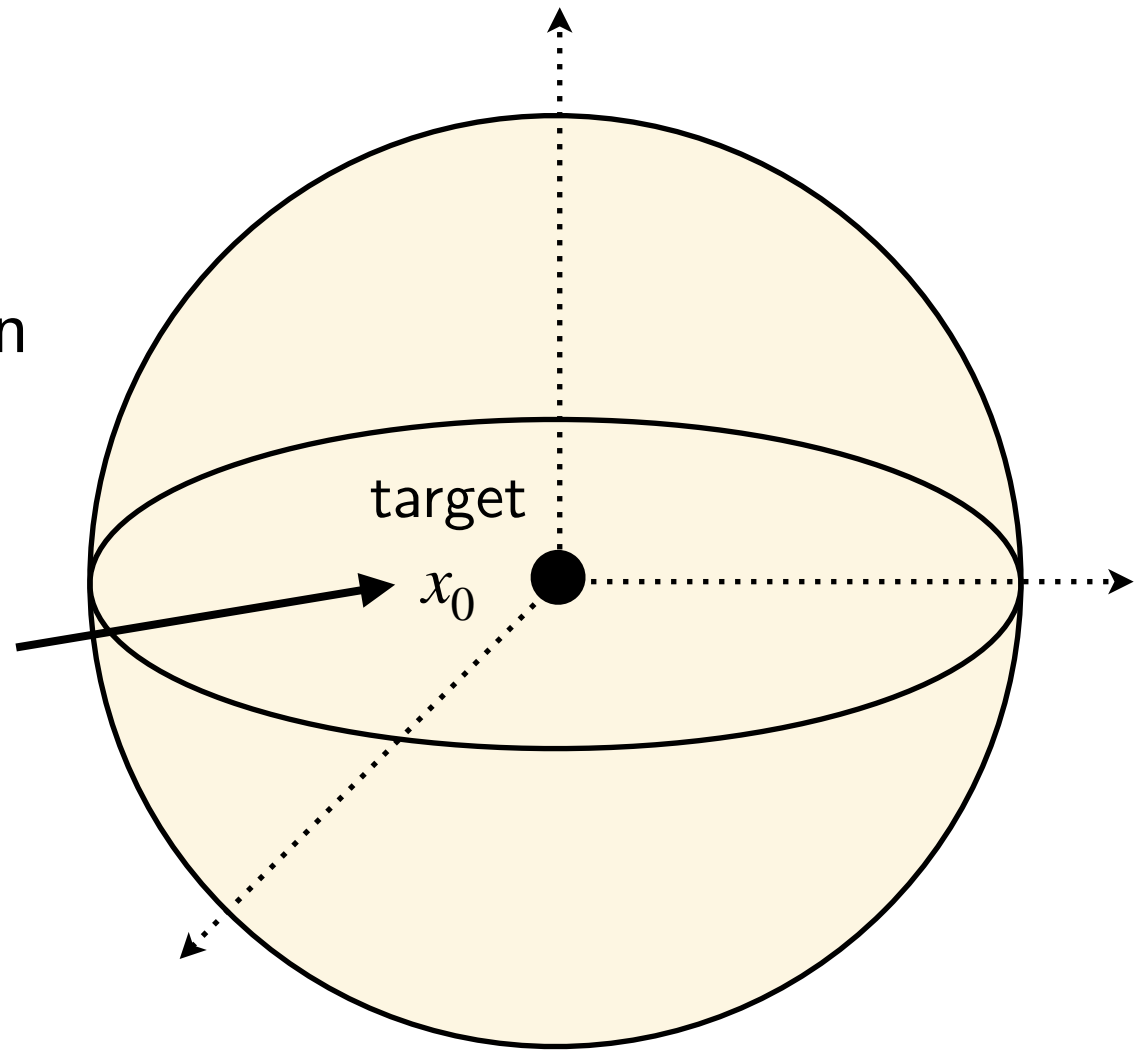


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The searcher's ideal match is normalized to the origin and denoted x_0 (the **target**)



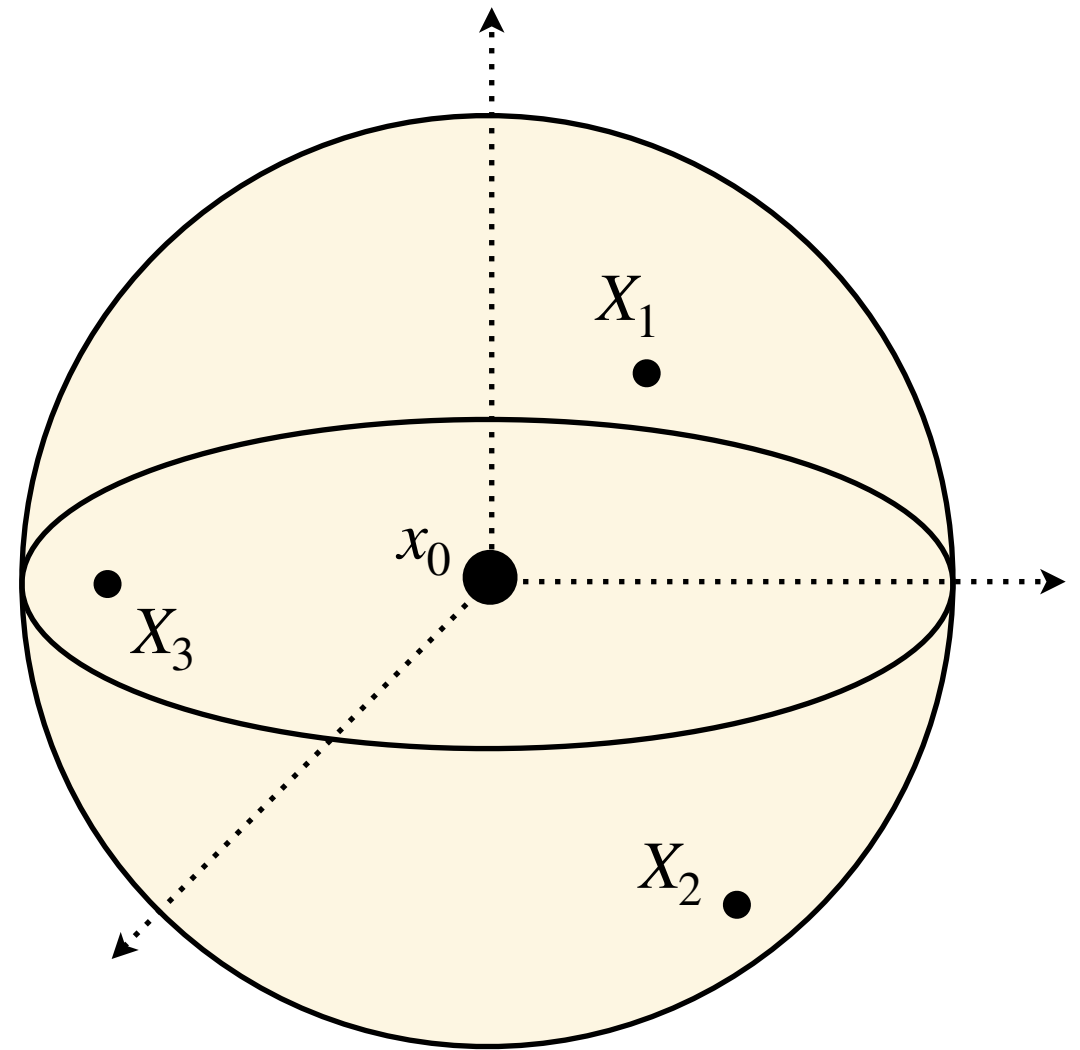
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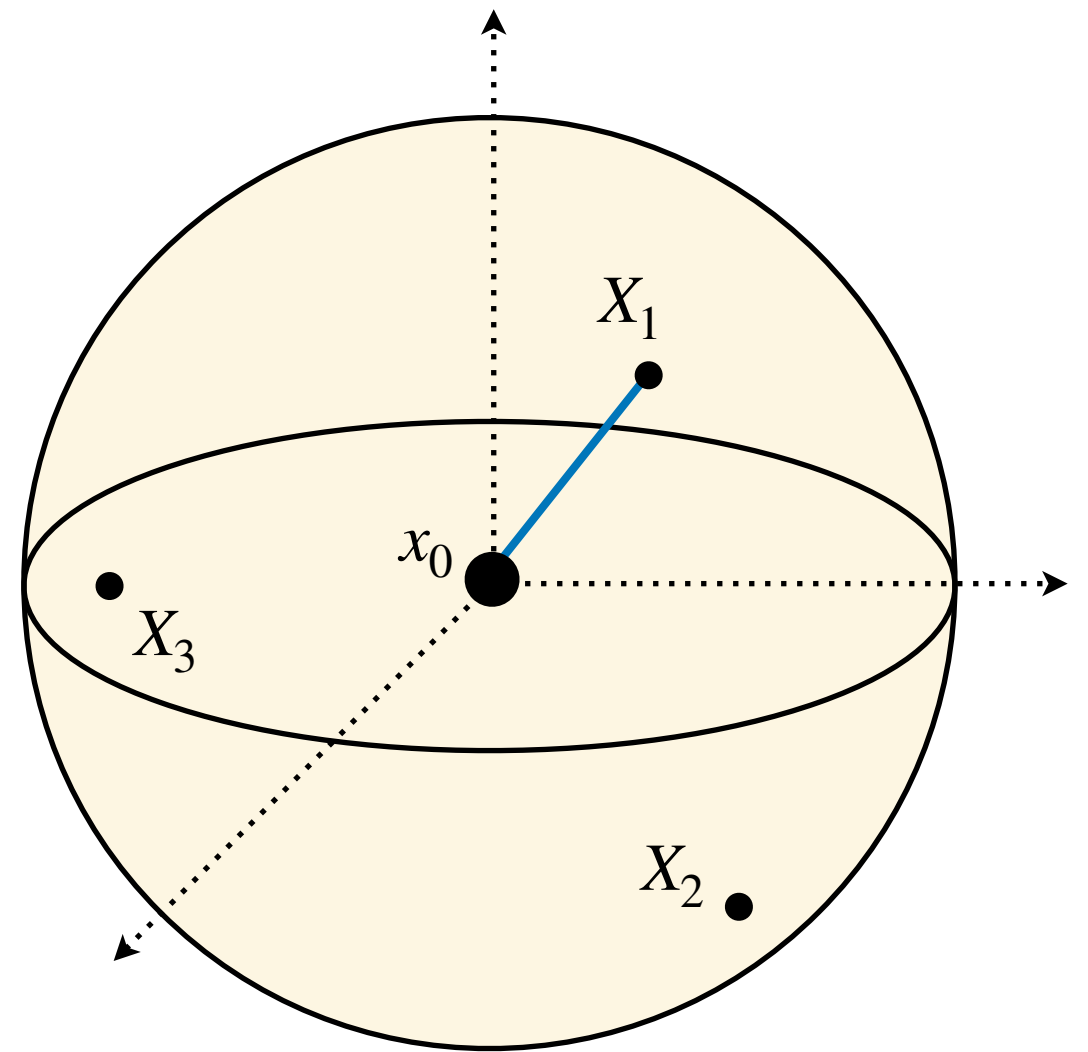
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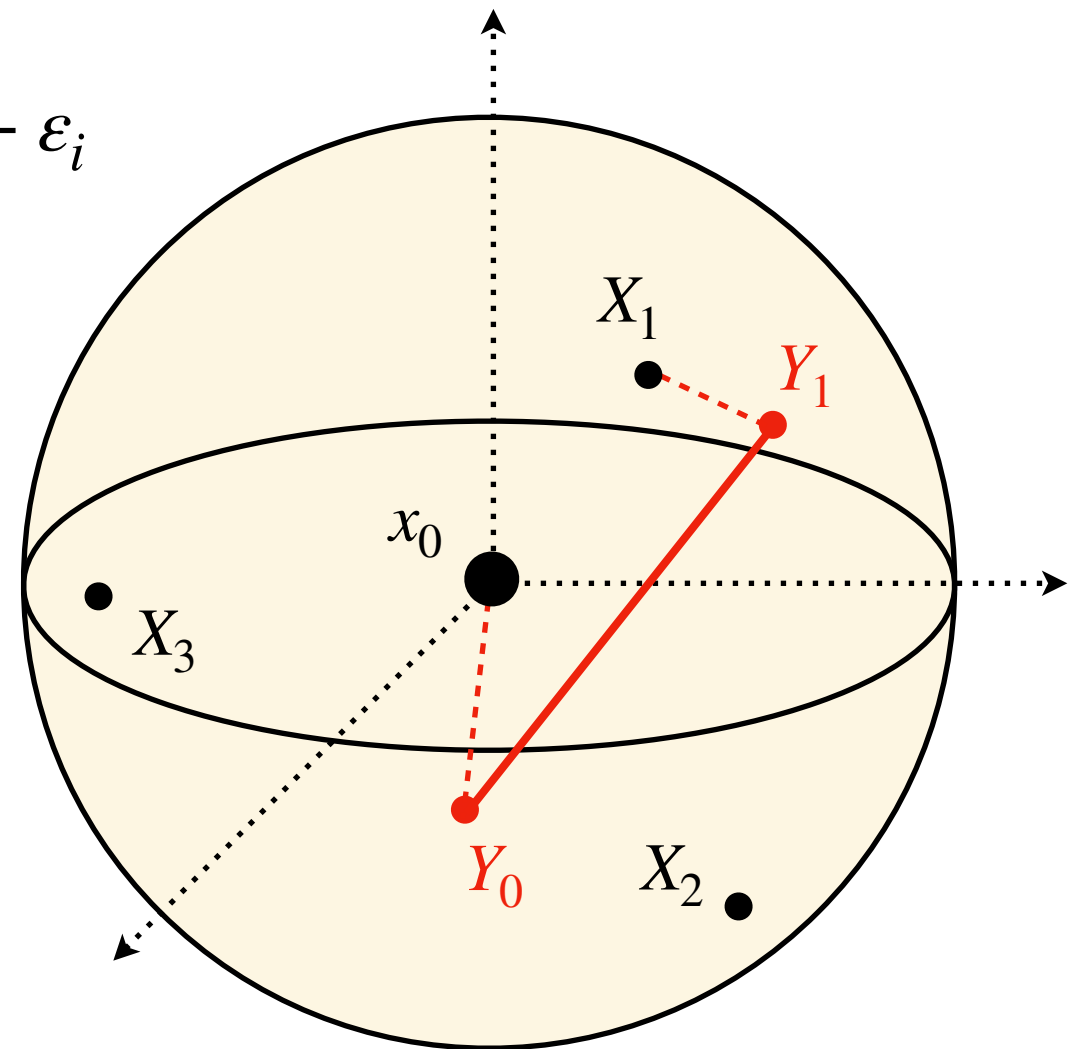
The searcher receives utility $-\|X_i\|$ from being matched to individual i



AI-Mediated Search

Under **AI-mediated** search,

- The searcher's target is represented by $Y_0 = x_0 + \varepsilon_0$
- Each individual i is represented by $Y_i = X_i + \varepsilon_i$
- The noise terms $\varepsilon_0, \varepsilon_i \sim \mathcal{N}(0, \sigma^2 I_k)$ are independent of one another
- Henceforth refer to Y_i and Y_0 as *AI representations*

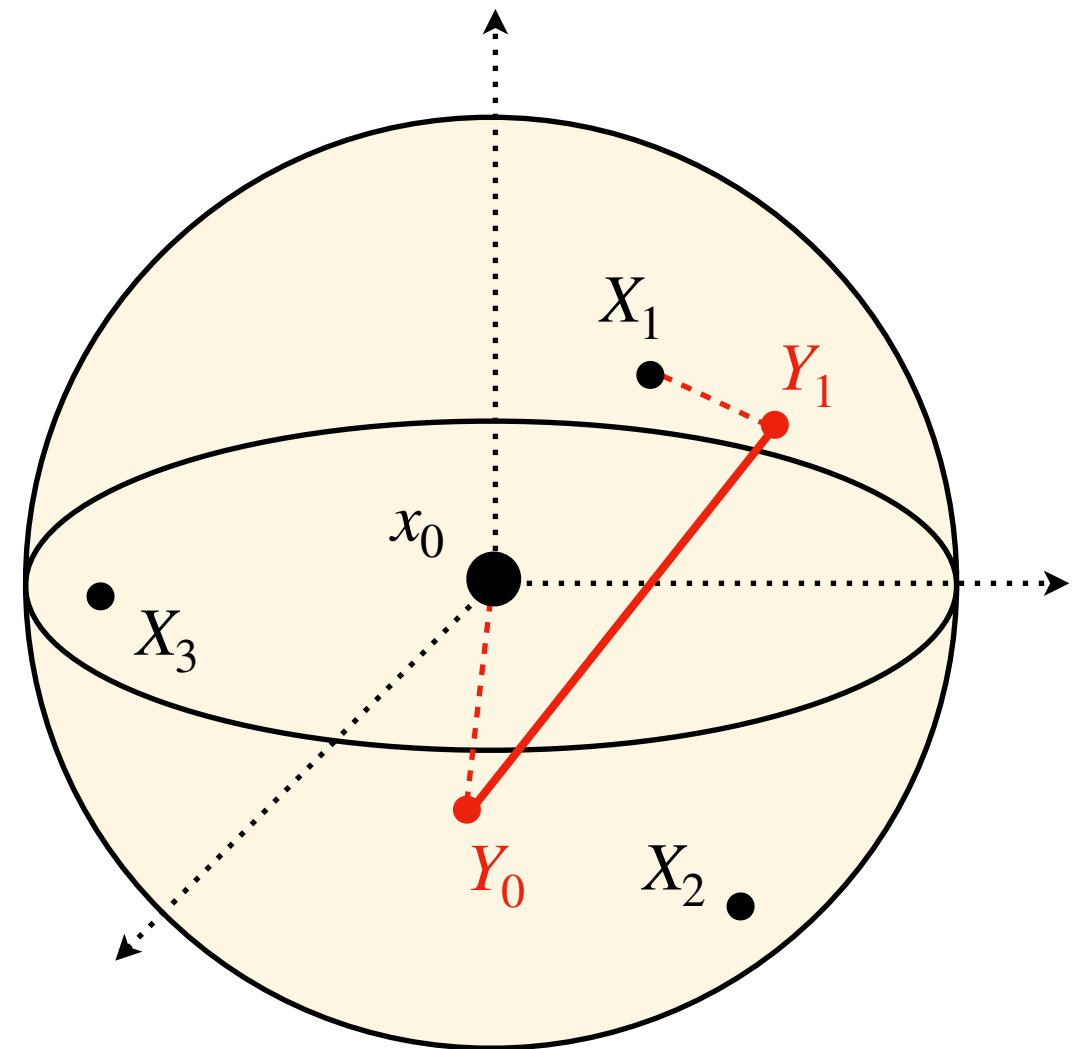


AI-Mediated Search

- The AI platform searches over n possible matches for the searcher
- The individual with the most compatible AI representation is

$$i_n^* = \operatorname{argmin}_{1 \leq i \leq n} \underbrace{\|Y_i - Y_0\|}_{\text{distance between the representation of the ideal point and the representation of individual } i}$$

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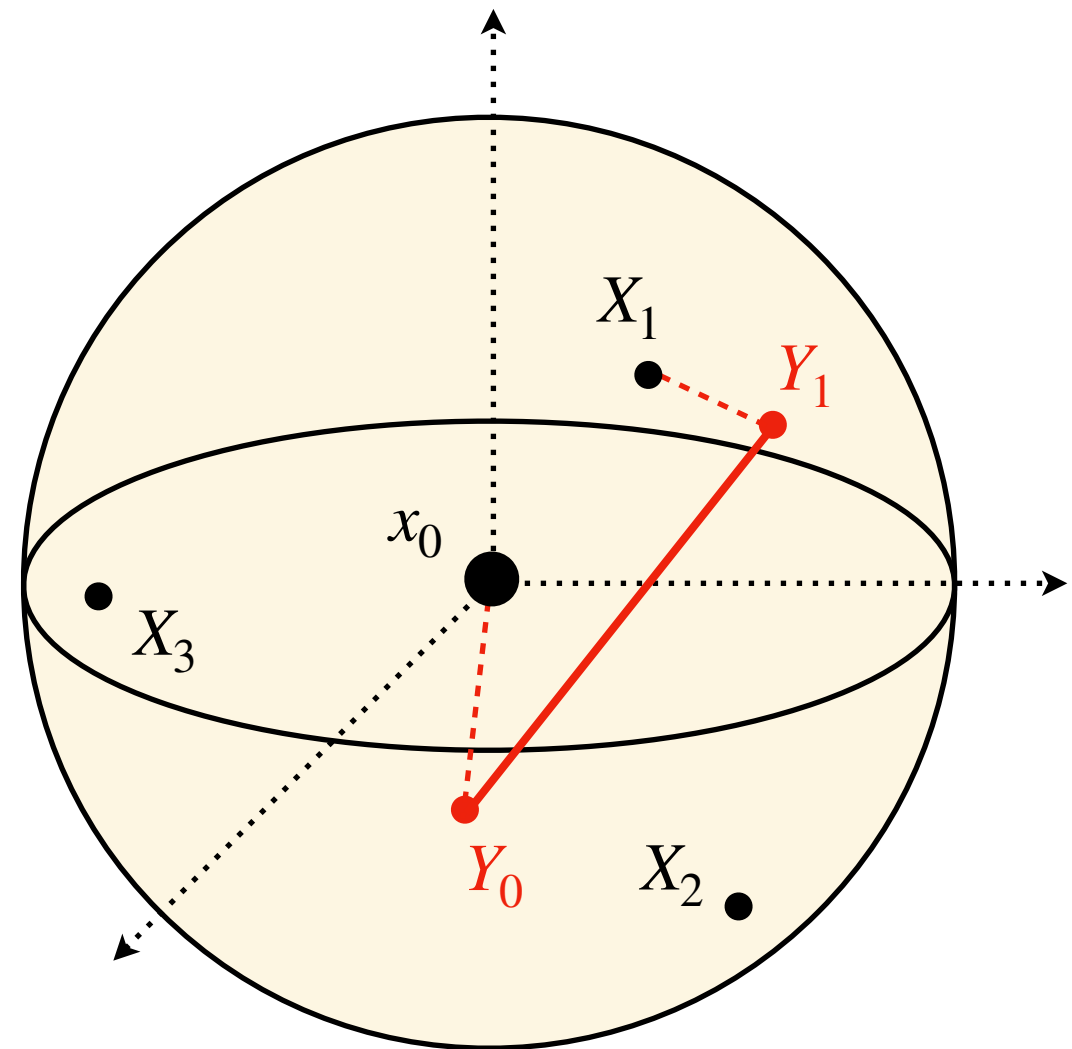
- The individual with the most compatible AI representation is

$$i_n^* = \operatorname{argmin}_{1 \leq i \leq n} \|Y_i - Y_0\|$$

- The searcher's expected utility is

$$u_k^{AI}(n) \equiv -E \left[\|X_{i_n^*}\| \right]$$

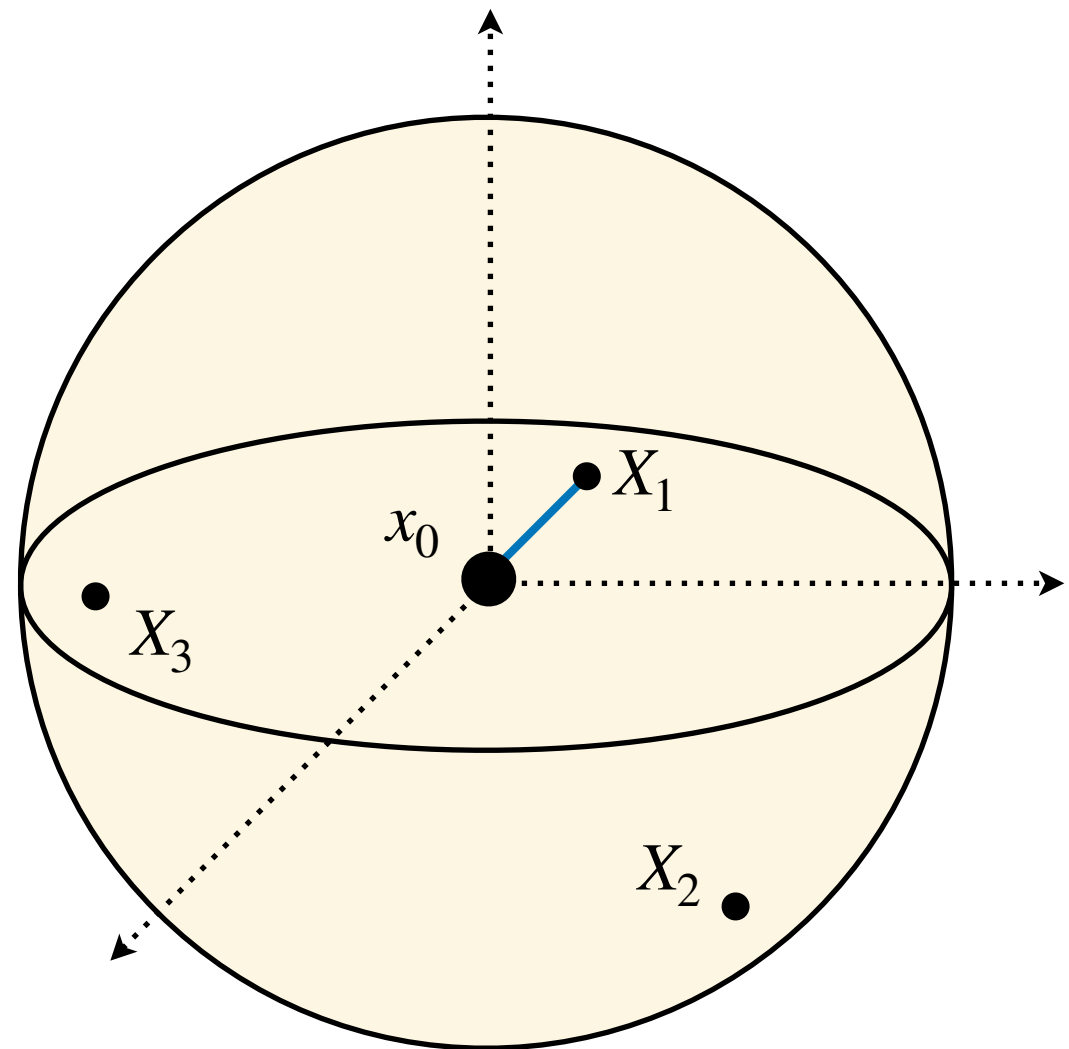
The selected match minimizes the distance between representations, but the searcher's payoff is based on the **true** distance between the underlying points



Benchmark: Best among m

- Suppose the searcher were to meet m individuals $X_1, \dots, X_m$ and match to the truly most compatible among them
- The searcher's expected utility from matching to this "best among m " is

$$u_k^B(m) \equiv - \mathbb{E} \left[\min_{1 \leq i \leq m} \|X_i\| \right]$$

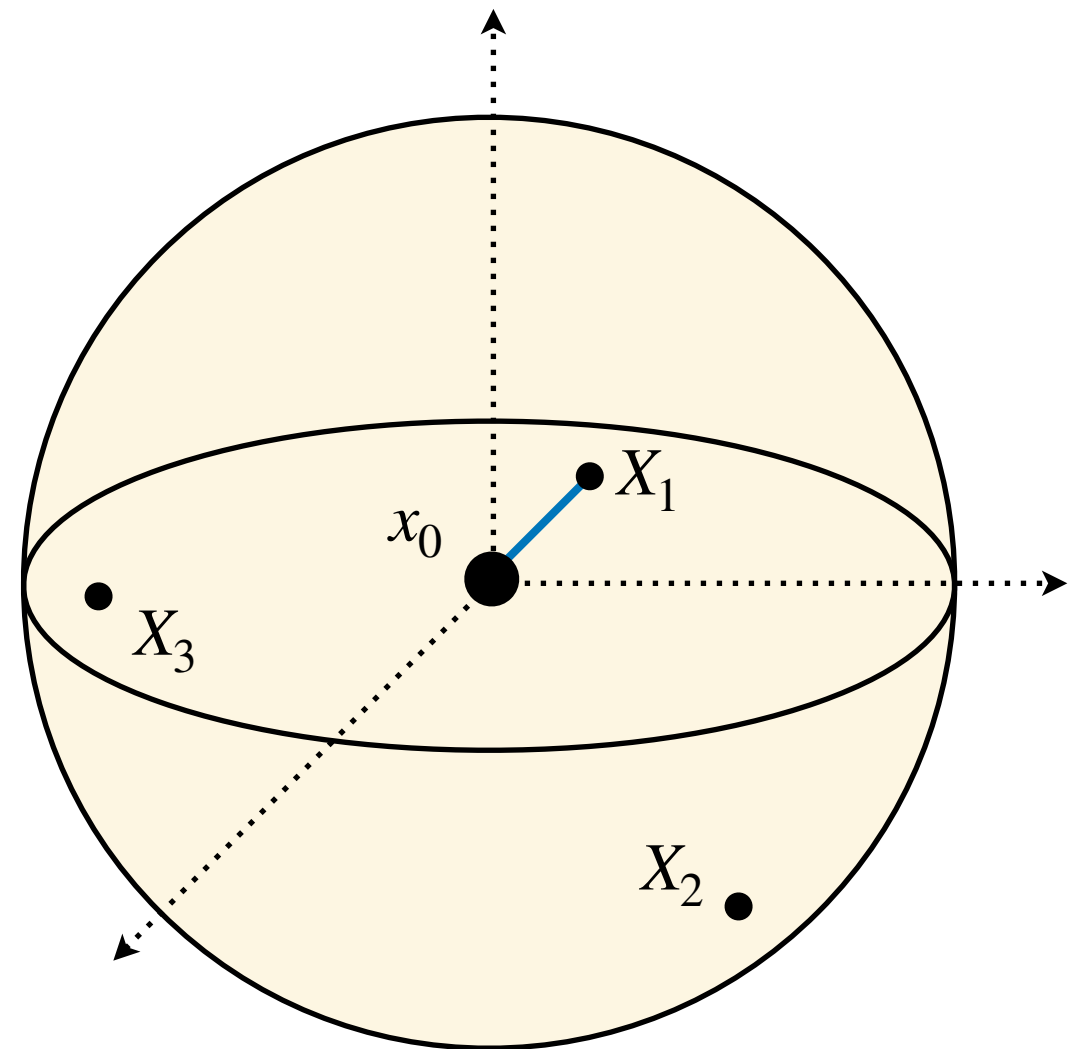


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- Can interpret as in-person search when humans have access to ground truth
- Or take this to be a theoretical benchmark



AI-Equivalent Sample Size

Measure of the value of AI-mediated search:

Definition: For any number of dimensions k , the *AI-equivalent sample size* m_k^* is the smallest $m \in \mathbb{Z}_+$ satisfying

$$u_k^B(m) > u_k^{AI}(n) \quad \forall n \in \mathbb{Z}_+$$

if the inequality above is not satisfied for any finite m , then $m_k^* = \infty$.

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expected utility of best match
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expected utility of best match
on platform with n candidates
(AI-mediated search)

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matching to the best candidate among m draws beats the AI platform, no matter how large it is

Results

Main Results

Proposition: For every number of dimensions $k \in \mathbb{Z}_+$, the AI-equivalent sample size m_k^* is finite

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Theorem: For all k sufficiently large, the AI-equivalent sample size is $m_k^* = 2$

- When there are sufficiently many relevant attributes, being on the platform is no better than meeting two people and matching to the better among them
- Underscores the limits of AI representations for complex search problems

Proof Idea/Intuition

Expected Utility:

Benchmark $u_k^B(m) \equiv - \mathbb{E} \left[\min_{1 \leq i \leq m} \|X_i\| \right]$

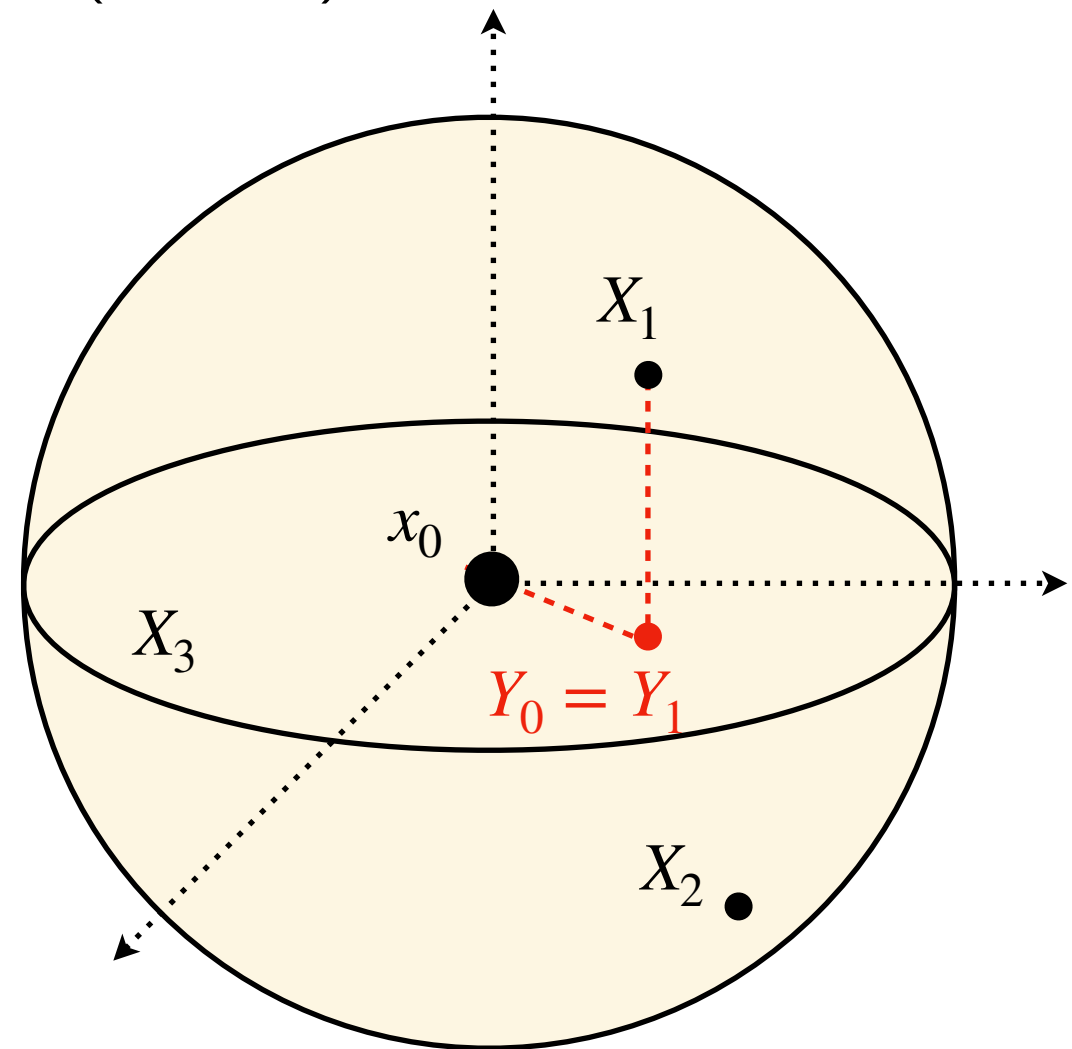
AI-mediated search $u_k^{AI}(n) \equiv - \mathbb{E} \left[\|X_{i_n^*}\| \right]$

need to show: $u_k^B(2) > u_k^{AI}(n)$ for all n

Proof Idea/Intuition

Define $u_k^{AI}(\infty) \equiv -E(\|X_i\| : \|Y_i - Y_0\| = 0)$

- The expected distance to an individual whose AI representation perfectly matches the target representation
- Can interpret as completely saturated limit ($n = \infty$)



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Next argue that $u_k^{AI}(\infty)$ converges to $u_k^B(1)$ as k grows large

- Getting your best match on the infinite AI platform is eventually as bad as matching to the first person you meet
- So matching to the best among **two** is better

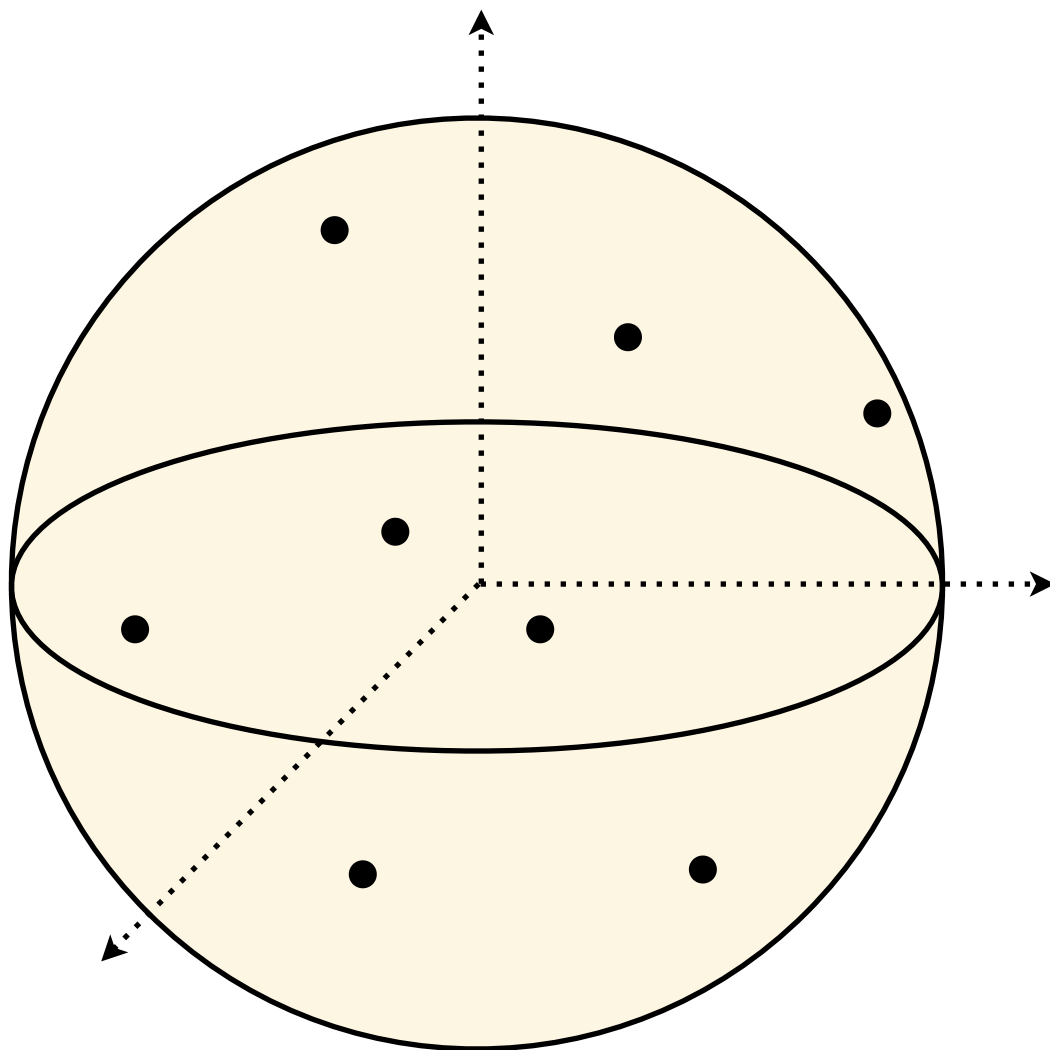
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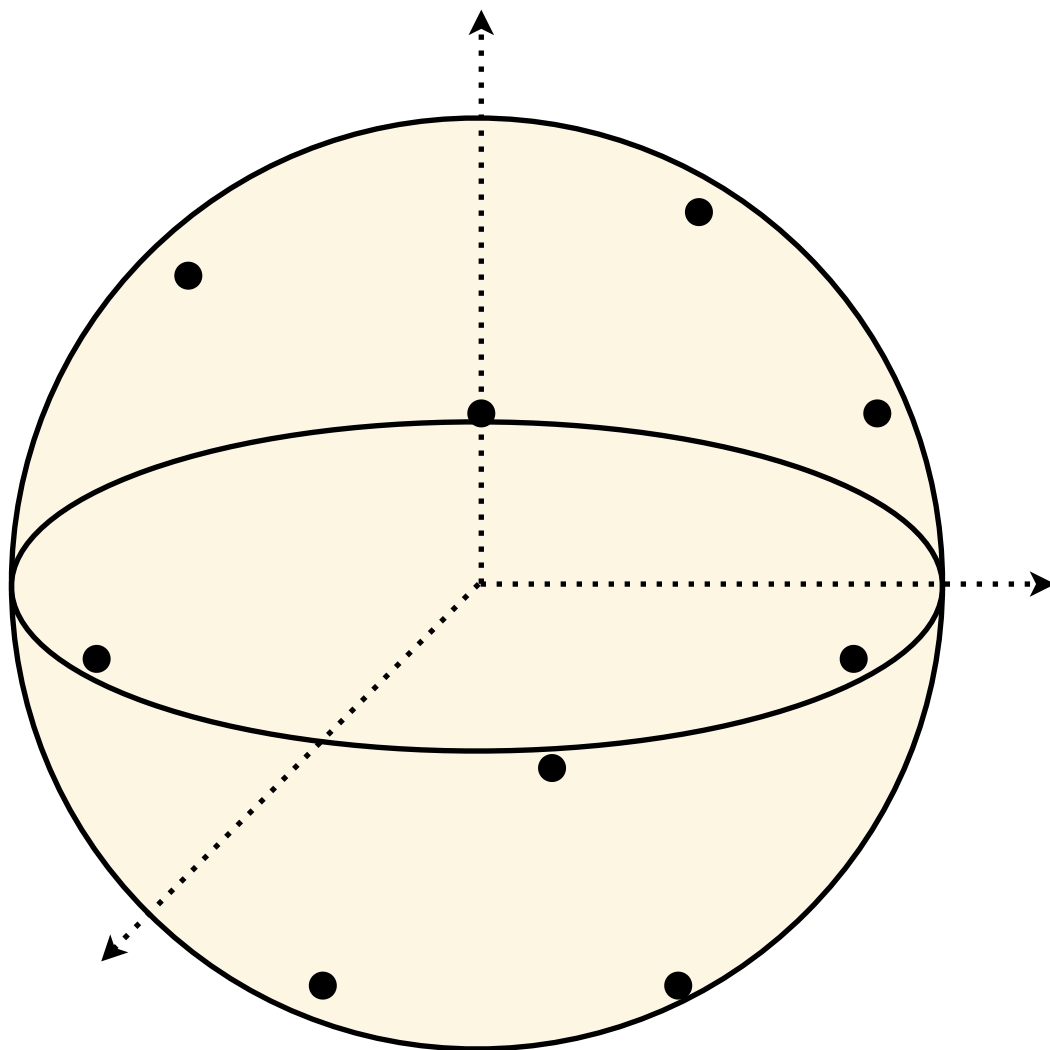
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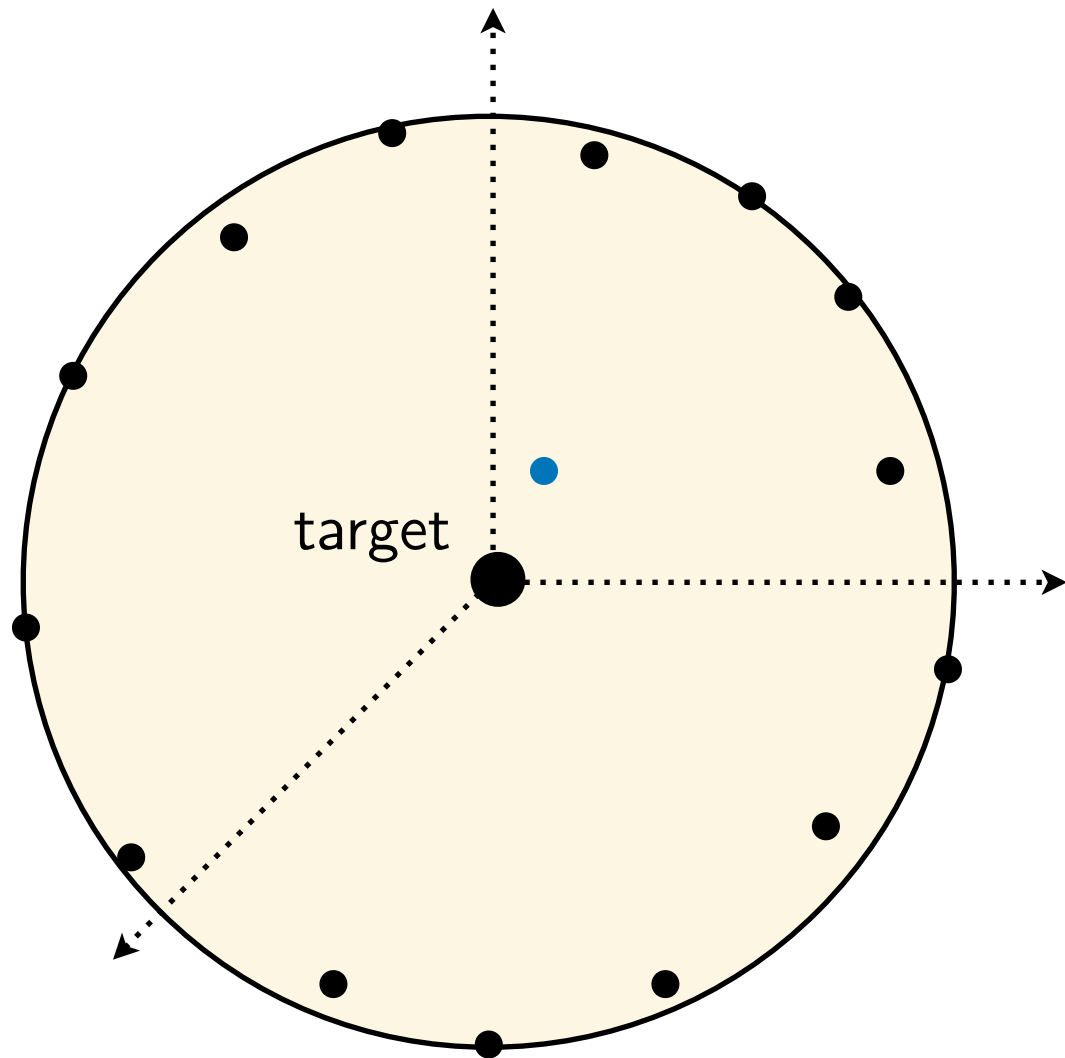
In fact, in high dimensions:

- points amass on the boundary of the unit ball
- distances between them concentrate around their expectations

i.e., people are increasingly unique and increasingly unlikely to exactly match the target

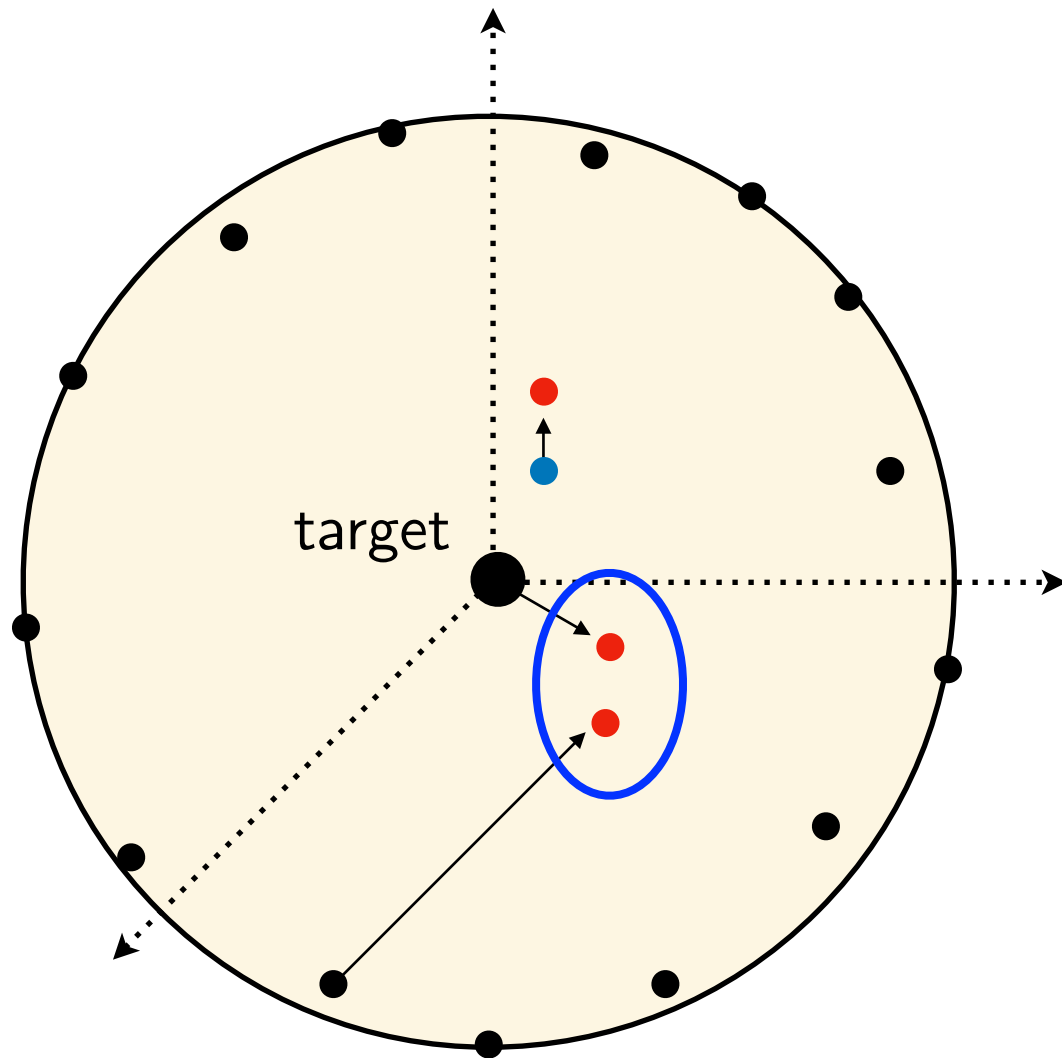
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- Best match increasingly driven by realization of the noise terms
- Eventually as if just selecting one candidate at random (i.e., $d_k^B(1)$)

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 - good spousal matches?
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- This is an empirical question

How Large is k ?

- In a simulation with $\sigma = 0.05$, the result holds for $k = 150$ dimensions
- How many dimensions of personality are relevant for identifying
 - good spousal matches?
 - good employees?
- This is an empirical question
- The main takeaway is that **the complexity of match compatibility matters** for how we should think about human vs. AI search
- The value of AI representations can differ across search problems depending on what the relevant k is

Extension

Extended Model

- In practice there is heterogeneity in the AI's ability to represent different individuals
- Many reasons for this
 - different quantities of personal data to share with AI (e.g., because of variation in social media usage)
 - different willingness to share it (e.g., because of variation in privacy preferences)
- Next suppose the population is split into a “data-rich” and “data-poor” group

Extended Model

- Let $g \in \{R, P\}$ denote the group (data-rich or data-poor)
- Groups are identical in characteristics: each X_i^g independently drawn from a uniform distribution on the unit ball
- Individuals i from group g have AI representations

$$Y_i^g = X_i^g + \varepsilon_i^g, \quad \varepsilon_i^g \sim \mathcal{N}(0, \sigma_g^2 I_k)$$

where $\sigma_R^2 < \sigma_P^2$ (lower error for data-rich group)

- The AI platform searches over n candidates from each group

Extended Model

Proposition: in the limit as the population grows large, the probability that the searcher's match is from the data-rich group **exceeds $1/2$**

- Although the two groups have identical compatibility with the searcher, a data-rich individual is more likely to be identified as the best match

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- Although the two groups have identical compatibility with the searcher, a data-rich individual is more likely to be identified as the best match

Proposition: in the limit as the number of dimensions and population size both grow large, the probability that the searcher's match is from the data-rich group **converges to 1**

Implications

- This systematic selection of data-rich individuals is not harmful to Subject:
 - The true distance between Subject and the selected individual is (in expectation) smaller if the individual is data-rich
 - Their AI clones more accurately represent them
- But this selection process has potentially concerning distributional implications
 - Individuals who are better understood by AI systems will be systematically preferred
 - This advantage could exacerbate existing inequalities
 - Moreover points to an emerging form of social stratification: an individual's opportunities may depend not only on their qualities, but also on how effectively AI systems can capture those qualities

Other discussion/extensions (see paper)

- role of parametric assumptions ➡
- two-sided matching ➡
- increasingly accurate AI ➡
- sequential search ➡
- human-AI integrated search ➡

Conclusion

- many papers have compared human and AI evaluation on tasks that are hard for both humans and machines
- but there are important evaluations that are intrinsically easier for people
 - no machine or expert knows better than ourselves whose company we enjoy
- in such settings, human evaluation is necessarily more accurate than the AI evaluation, but also more costly
- can automated search over a sufficiently large number of candidates compensate for the AI's errors during evaluation?
- this paper suggests that the answer depends on the complexity of the target: when sufficiently many attributes are relevant, then the answer is **no**

Thank You

role of parametric assumptions

- personalities are uniformly distributed and AI errors follow a multivariate Gaussian distribution
- proof of **theorem 1** ($m_k^* = 2$ for large k) relies heavily on these assumptions, especially the rate of convergence lemmas
- but key statistical phenomenon is more general: in high dimensions,
 - random points isolate from one another
 - the distances between them concentrate around their expectations
- expect generalizations to other distributions (in particular, normal-normal)



two-sided matching

- this paper's model of search is one-sided
- abstracts from the question of whether Subject's best match (once identified) would agree to this pairing
- I do not pursue a complete matching model
- but in many such models, Subject's expected payoff would decrease in the number of candidates considered by Subject's potential partners
- this would further favor in-person search, where each individual searches over fewer candidates



increasingly accurate AI

- main results hold for any quality of AI approximation (i.e., any $\sigma > 0$).
- but would expect that as the AI representations become more accurate, AI-mediated search becomes more (relatively) attractive
- I show that for every fixed number of dimensions k ,

$$d_{k,\sigma}^{AI}(\infty) \rightarrow 0 \text{ as } \sigma \rightarrow 0$$

- i.e., Subject's expected distance to best match on the AI platform converges to zero as the AI's approximation errors vanish
- thus, while the value to increasing the size of the AI platform is capped, the value to increasing the accuracy of the AI clones is not



sequential search

have so far considered a static search problem

consider the following model of sequential search in discrete time

- at $t = 0$, Subject chooses between the in-person regime and AI regime
- in the **in-person regime**, observe X_t in each period $t \geq 1$ and decide whether to stop (and match with best so far) or sample again at cost c_{IP}
- in the **AI regime**,
 - pay an upfront cost κ to participate on the platform
 - in each $t \geq 1$, observe clone distance $\|Y_t - Y_0\|$ and decide whether to stop (and match with closest clone) or sample again at cost c_{AI}

result: Subject prefers the in-person regime when $\kappa > 2c_{IP}$ (i.e., the fixed cost of joining the platform exceeds the cost of meeting two people)



human-AI integration

- one might consider integrating the two approaches as follows:
 - the AI platform simulates interactions between Subject and n individuals using their AI clones
 - recommends to Subject those m candidates whose AI clones are closest to Subject's clone (previously just one)
 - Subject meets these individuals in person and matches to the most compatible among them
- this is **unlikely** to substantially improve Subject's payoffs



human-AI integration

- previously showed that the best match on the AI platform is (for large k) similar to a draw from a uniform distribution over the unit ball
- conjecture that the m best matches on the AI platform are also (for large k) similar to m draws from a uniform distribution on the unit ball
- thus choosing among the AI-sourced candidates is not substantially better than choosing from an equal number of randomly drawn candidates
- suggests that:
 - for any fixed platform entry cost, Subject should optimally choose not to participate on the platform when k is large
 - for large k , Subject should prefer searching over $m + 1$ people in person rather than selecting from m people sourced by the AI platform



intuition

- recall that the selected match i minimizes $\|Y_i^g - Y_0^g\|$
- in a large population, this is an individual for whom that distance is very small
- which group is that extreme draw more likely to come from?
- the smallest distance to any group- g individual, i.e.,

$$M_n^g = \min_{1 \leq i \leq n} \|Y_i^g - Y_0^g\|$$

is asymptotically distributed like an exponential random variable with rate proportional to $f_k^g(0)$ (the density of $Y_i^g - Y_0^g$ at the zero vector)

- so group g 's probability of “winning” is asymptotically determined by the relative size of its exponential rate, and thus by the size of $f_k^g(0)$
- finally, lower noise variance leads to a more sharply peaked density at zero, and hence a higher probability of being chosen